
Basketball Prediction



Machine Learning Class Project

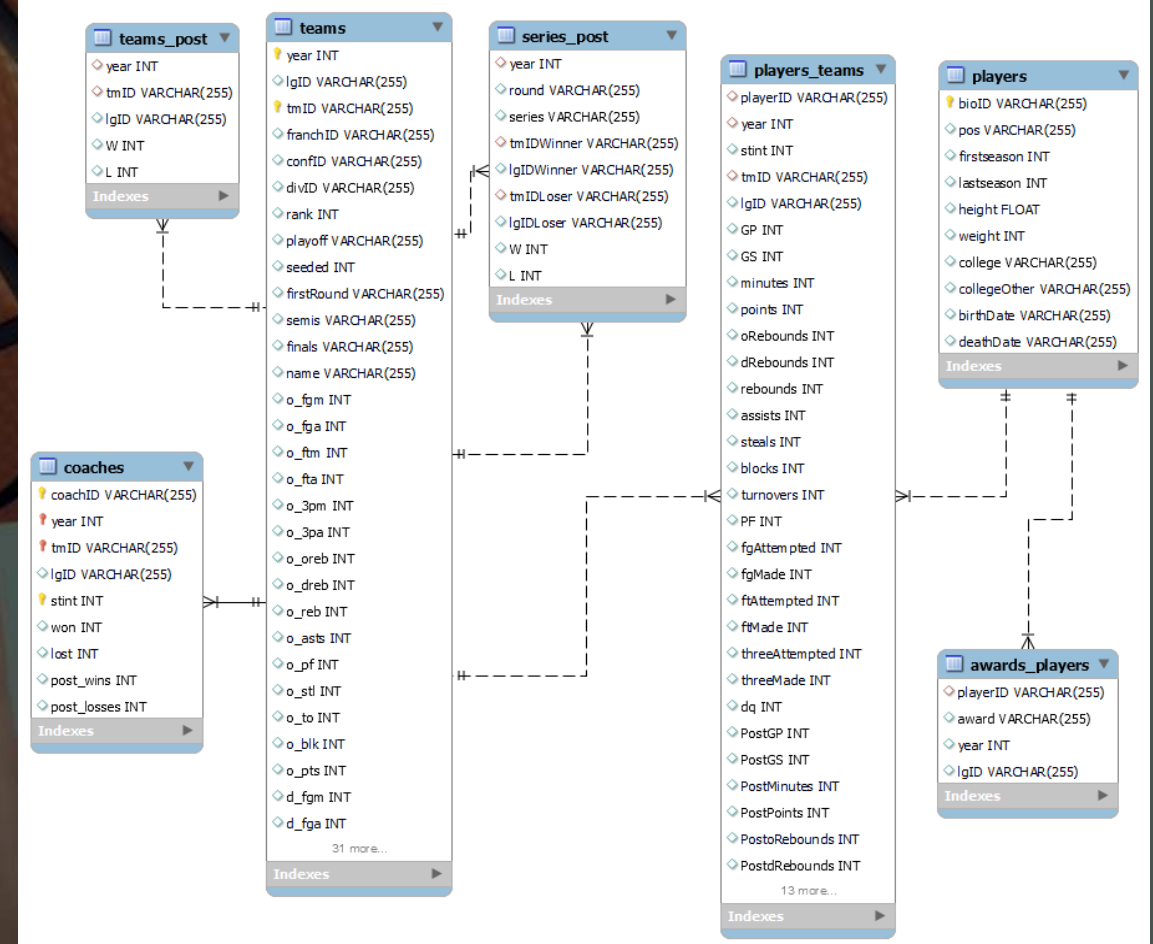
- Anu Atolagbe (up202400090) [IF: 0.40]
 - Rui Silveira (up202108878) [IF: 0.85]
 - Tomás Vicente (up202108717) [IF: 1.00]
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Project Domain

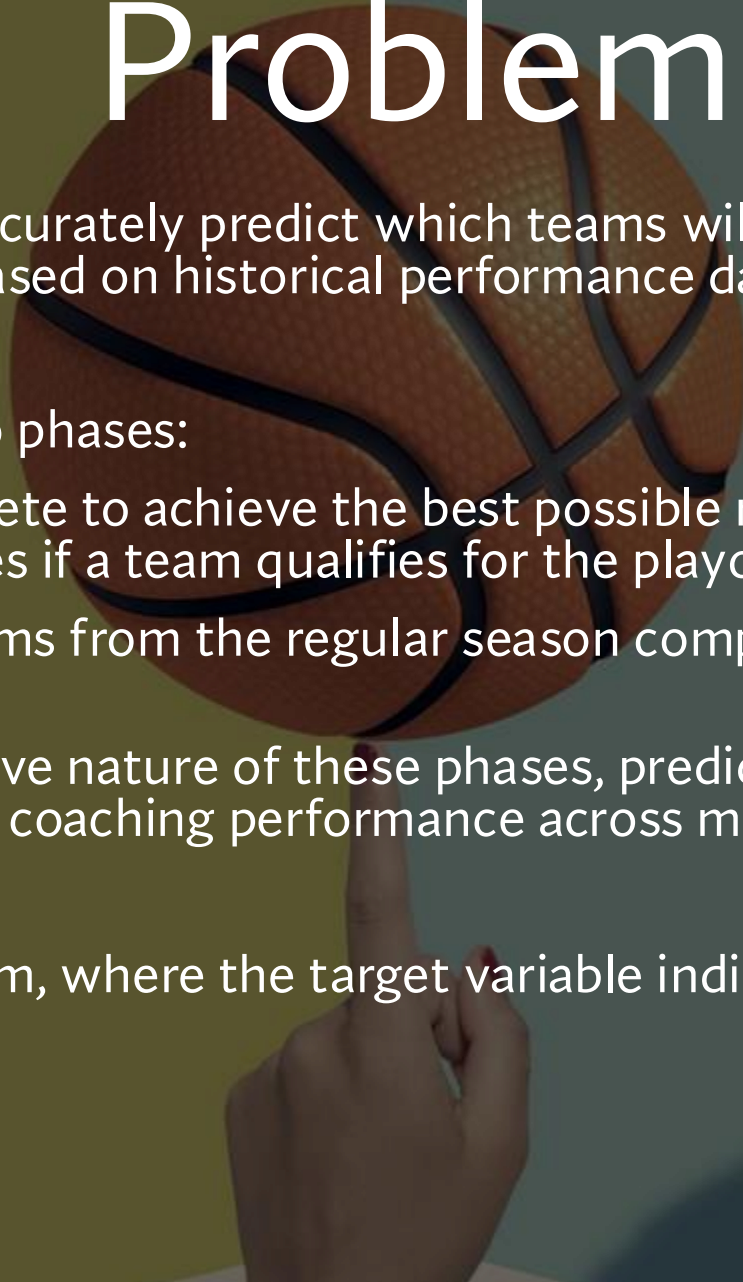
Dataset Description

This project leverages a comprehensive dataset spanning a decade, which includes:

- **Teams:** 142 records representing 20 teams.
- **Players:** 893 records detailing individual player statistics.
- **Coaches:** 162 records across 57 coaches, capturing their impact on team performance.
- **Awards_Players:** 95 records documenting achievements such as MVPs and All-Star recognitions.
- Some other tables connecting the team's or player's performance



Problem Definition

A basketball is balanced on the tip of a finger. The background is a gradient of dark green and blue.

Objective

The objective of this project is to accurately predict which teams will qualify for the playoffs in the next season of the WNBA, based on historical performance data.

Key Problem

Each WNBA season consists of two phases:

- **Regular Season:** All teams compete to achieve the best possible record. The number of wins during this phase determines if a team qualifies for the playoffs.
- **Playoff Phase:** The top eight teams from the regular season compete in elimination rounds to determine the champion.

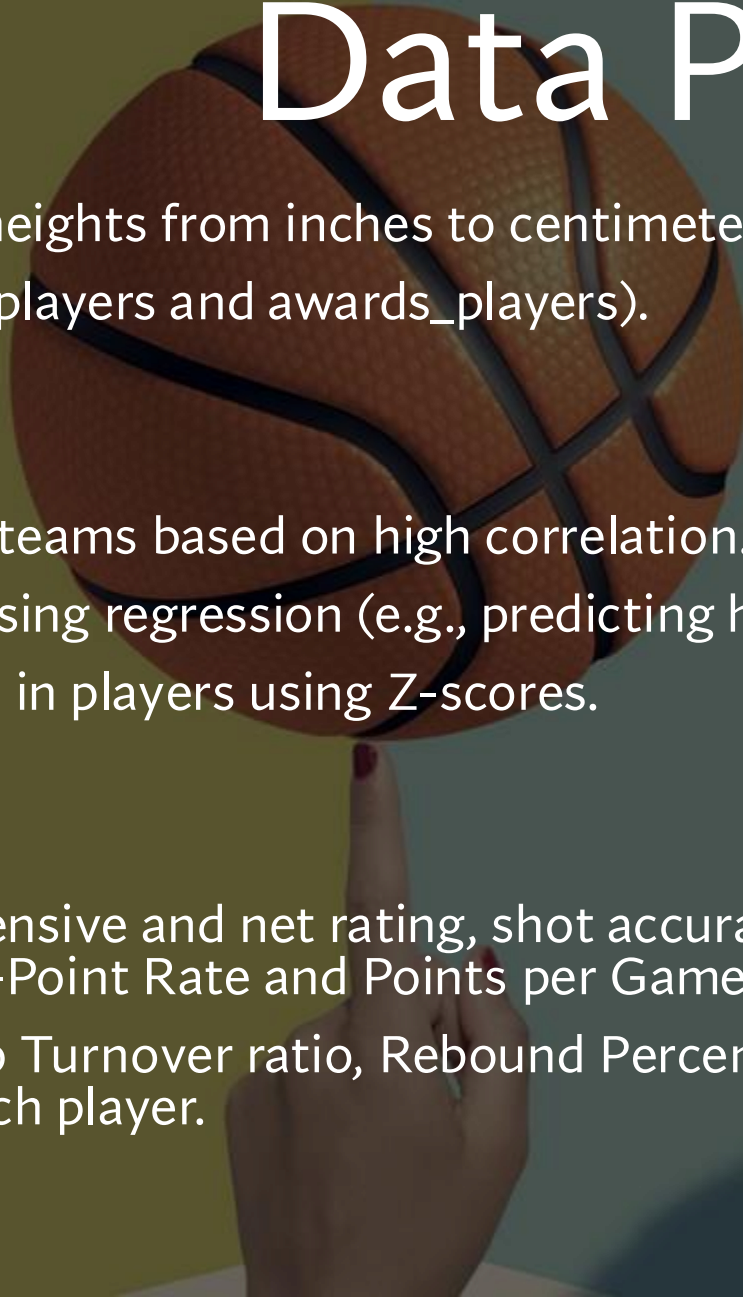
Given the complexity and competitive nature of these phases, predicting playoff qualification requires analyzing team, player, and coaching performance across multiple seasons.

Approach

This is a binary classification problem, where the target variable indicates whether a team qualifies for the playoffs:

- **Playoff:** Yes (1) or No (0).

Data Preparation

A hand with a red-painted index finger is balancing a basketball on its tip. The background is a dark, textured grey.

Data Integration:

- Converted formats (e.g., player heights from inches to centimeters).
- Matched bioID across datasets (players and awards_players).

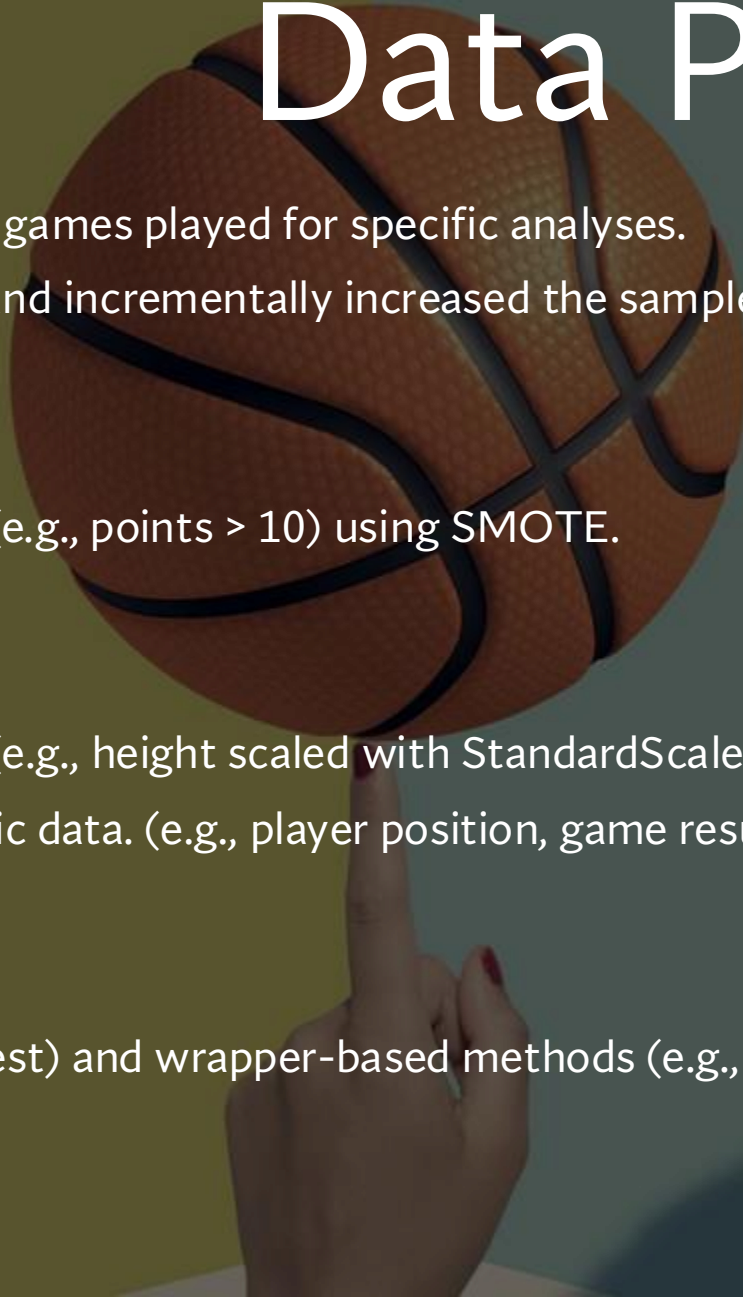
Data Cleaning:

- Removed redundant columns in teams based on high correlation.
- Filled missing values in players using regression (e.g., predicting height from weight).
- Identified and addressed outliers in players using Z-scores.

Feature Engineering:

- Created win_rate, offensive, defensive and net rating, shot accuracy, Offensive, Defensive and Overall Rebound Percentages, 3-Point Rate and Points per Game for teams.
- Created Shot Accuracy, Assist to Turnover ratio, Rebound Percentages, Steal Rates, Block Rates, Points per Game and PER for each player.

Data Preparation

A basketball is balanced on the tip of a finger. The background is a gradient of dark green and blue.

Sampling:

- Focused on players with more than 50 games played for specific analyses.
- Started with small subsets of players and incrementally increased the sample size for development.

Handling Imbalanced Data:

- Balanced binary classification targets (e.g., points > 10) using SMOTE.

Data Transformation:

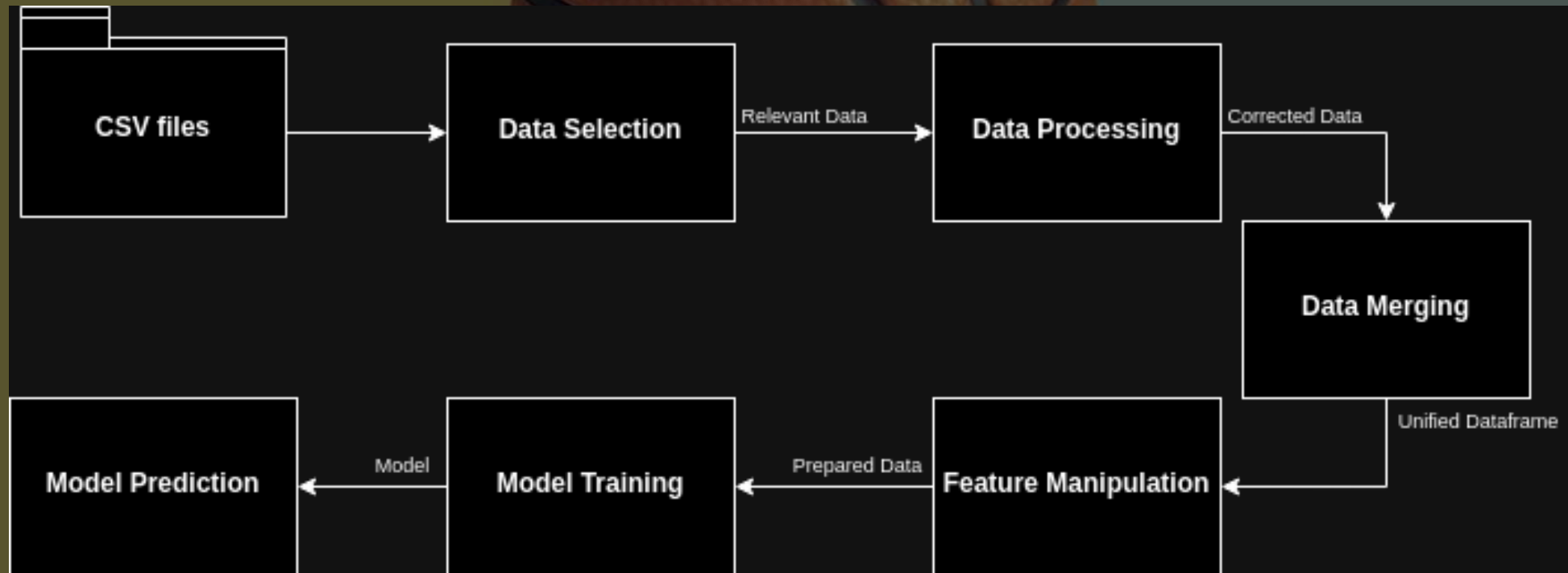
- Applied scaling to numerical features (e.g., height scaled with StandardScaler).
- Transformed nominal data into numeric data. (e.g., player position, game results (finals, semis, ...))

Feature Selection:

- Combined filter-based (e.g., SelectKBest) and wrapper-based methods (e.g., RFE) to select top features.

Experimental Setup

Pipeline



Results

Model Evaluation

We have tried various predictive models, like Decision Trees, Random Forest, Logistic Regression and Support Vector Regression.

We have tried training the models by feeding it just the previous year data and the whole previous years data.

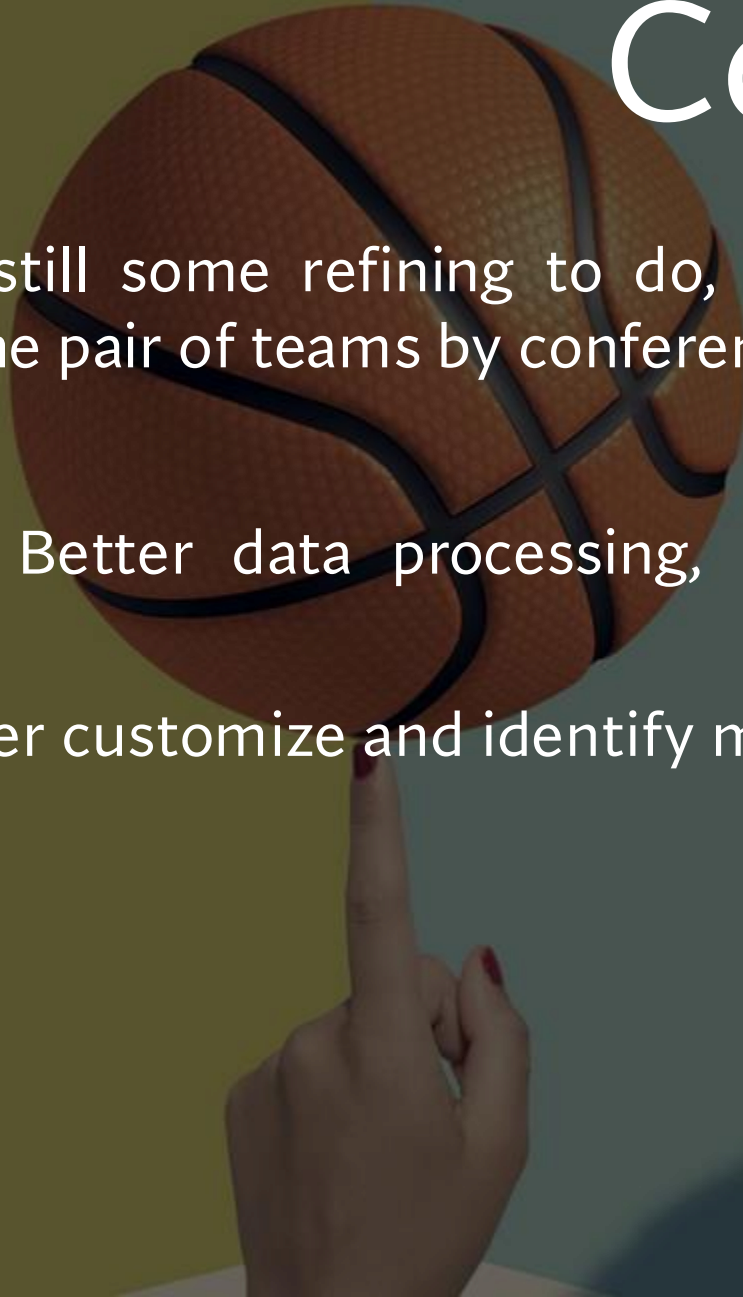
Later only focusing in the Random Forest model, as it stood out to be the most effective.

The formulas used for evaluation metrics:

- Acc: $TP / (\#cases)$
- Error: $\text{sum}(|\text{guess} - \text{label}|) / \text{MaxError}$

Model (acc, error)	Single Year Training	Iteractive Training
Decision Tree Classifier	(0.53, 0.56)	(0.51, 0.58)
Random Forest Classifier	(0.65, 0.44)	(0.66, 0.43)
Logistic Regression	(0.64, 0.39)	(0.53, 0.47)
SVR	(0.62, 0.44)	(0.60, 0.44)

Conclusion

A hand with red-painted fingernails is shown from the wrist up, balancing a brown basketball on its index finger. The background is a dark, muted green with a subtle gradient.

Conclusion

Even though our work has still some refining to do, we are getting good results, just missevaluating one pair of teams by conference.

Future Work

- Further Data Cleaning – Better data processing, outlier identification, missing parameters, etc.
- Feature Engineering - Better customize and identify most relevant features to the model.

Annexes

Data Preparation

```
Index(['year', 'tmID', 'confID', 'rank', 'firstRound', 'semis', 'finals',  
      'o_fgm', 'o_ftm', 'o_dreb', 'o_reb', 'o_ast', 'o_pf', 'o_stl', 'o_blk',  
      'd_fgm', 'd_ftm', 'd_3pm', 'd_3pa', 'd_reb', 'd_ast', 'd_pf', 'd_stl',  
      'd_blk', 'won_x', 'lost_x', 'GP', 'confW', 'confL', 'win_loss_ratio',  
      'ORtg', 'DRtg', 'NRTg', 'TSht_x', 'OREb', 'DREb', 'Reb', '3PtR',  
      'PPG_x', 'playoff_qualification', 'W', 'L', 'points', 'rebounds',  
      'assists', 'steals', 'blocks', 'turnovers', 'TSht_y', 'AstTO', 'RebP',  
      'OREbP', 'DREbP', 'StlR', 'BlkR', 'PPG_y', 'num_awards', 'won_y',  
      'lost_y', 'post_wins', 'post_losses'],  
      dtype='object')
```

Updated Columns in the data fed to the model

Annexes

Data Preparation

Before conversion:

	bioID	height
0	abrahta01w	74.0
1	abrossv01w	74.0
2	adairje01w	76.0
3	adamsda01w	73.0
4	adamsjo01w	75.0

After conversion:

	bioID	height_cm
0	abrahta01w	187.96
1	abrossv01w	187.96
2	adairje01w	193.04
3	adamsda01w	185.42
4	adamsjo01w	190.50

Players heights from inches to centimeters

Annexes

Model Evaluation

```
Year 1 -> 2:
Results:
predict: [0.82 0.82 0.47 0.75 0.18 0.84 0.78 0.15 0.73 0.08 0.22 0.05 0.83 0.31
0.81 0.18]
label:   [1. 1. 0. 1. 0. 1. 1. 0. 1. 0. 0. 0. 1. 0. 1. 0.]
expected: [1. 0. 0. 1. 1. 1. 0. 0. 1. 0. 0. 0. 0. 1. 1. 1.]
error:    [0.18 0.82 0.47 0.25 0.82 0.16 0.78 0.15 0.27 0.08 0.22 0.05 0.83 0.69
0.19 0.82]
Accuracy: 0.62
Error:    6.78 / 16 (0.42)

Year 2 -> 3:
Results:
predict: [0.89 0.87 0.27 0.65 0.84 0.18 0.39 0.17 0.81 0.09 0.63 0.15 0.4 0.15
0.62 0.89]
label:   [1. 1. 0. 1. 1. 0. 0. 0. 1. 0. 1. 0. 0. 0. 1. 1.]
expected: [1. 1. 1. 1. 0. 1. 0. 1. 0. 0. 0. 0. 0. 1. 0. 0.]
error:    [0.11 0.13 0.73 0.35 0.84 0.82 0.39 0.83 0.81 0.09 0.63 0.15 0.6 0.15
0.62 0.89]
Accuracy: 0.44
Error:    8.14 / 16 (0.51)

Year 3 -> 4:
Results:
predict: [0.64 0.62 0.66 0.66 0.67 0.59 0.66 0.66 0.66 0.06 0.15 0.67 0.67 0.63]
label:   [0. 0. 1. 1. 1. 0. 1. 1. 1. 0. 0. 1. 1. 0.]
expected: [0. 0. 1. 1. 0. 0. 1. 1. 1. 0. 1. 0. 1. 1.]
error:    [0.64 0.62 0.34 0.34 0.67 0.59 0.34 0.34 0.34 0.06 0.85 0.67 0.33 0.37]
Accuracy: 0.71
Error:    6.5 / 14 (0.46)
```

Snippet of year by year model evaluation

```
Accuracy [0.62, 0.44, 0.71, 0.69, 0.85, 0.57, 0.54, 0.71, 0.38]
Error     [0.42, 0.51, 0.46, 0.32, 0.28, 0.44, 0.4, 0.34, 0.61]
```

Average Performance Over All Years:

Average Accuracy: 0.64

Average Error: 0.4

Snippet of overall model evaluation

Annexes

Model Evaluation (Updated)

```
Year 6 → 7:  
Results:  
predict:  
[0.03 0.14 1.05 0.64 0.89 1.03 0.68 0.04 0.96 0.35 1.05 0.05 0.93 0.17]  
label:      [0. 0. 1. 1. 1. 1. 1. 0. 1. 0. 1. 0. 1. 0.]  
expected: [0. 0. 1. 1. 1. 1. 1. 0. 1. 0. 1. 0. 1. 0.]  
error:      [0.03 0.14 0.05 0.36 0.11 0.03 0.32 0.04 0.04 0.35 0.05 0.05 0.07 0.17]  
Accuracy: 1.0  
Error:      1.81 / 14
```

```
Year 7 → 8:  
Results:  
predict:  
[0.03 0.98 0.97 0.92 0.9 0.89 0.08 0.05 0.38 0.9 0.2 0.81 0.91]  
label:      [0. 1. 1. 1. 1. 1. 0. 0. 0. 1. 0. 1. 1.]  
expected: [0. 1. 1. 1. 1. 1. 0. 0. 0. 1. 0. 1. 1.]  
error:      [0.03 0.02 0.03 0.08 0.1 0.11 0.08 0.05 0.38 0.1 0.2 0.19 0.09]  
Accuracy: 1.0  
Error:      1.46 / 13
```

```
Accuracy [1.0, 0.93, 0.92, 1.0, 1.0, 1.0, 1.0, 1.0]  
Error    [4.73, 1.76, 3.07, 2.22, 1.81, 1.46, 1.27, 1.04]
```

Average Performance Over All Years:

Average Accuracy: 0.98

Average Error: 2.17

Snippet of year by year model evaluation

Snippet of overall model evaluation

Annexes

Model Evaluation

Window Span	1	2	3	4	5	6	7	8	9
(acc, error)	(0.65, 0.44)	(0.64, 0.43)	(0.66, 0.43)	(0.62, 0.44)	(0.64, 0.43)	(0.66, 0.43)	(0.66, 0.43)	(0.66, 0.43)	(0.66, 0.43)

Window Span	1	2	3	4	5	6	7	8	9
(acc, error)	(0.64, 0.39)	(0.59, 0.44)	(0.58, 0.48)	(0.57, 0.49)	(0.53, 0.47)	(0.55, 0.47)	(0.53, 0.47)	(0.53, 0.47)	(0.53, 0.47)

Year Window Span exploration and results on Random Forest and Logistic Regression models (respectively)

Annexes

Model Feature Importances

1	feature,importance
2	rank,0.1737819300784389
3	L,0.13040941958313643
4	win_loss_ratio,0.09722032081509631
5	won_x,0.09314052527920612
6	confW,0.08975766143747817
7	confL,0.07811468038908782
8	lost_x,0.06914043900278173
9	ORtg,0.03328437629508119
10	DRtg,0.023099100164871295
11	W,0.021724255364756962
12	lost_y,0.016835135663611958
13	NRtg,0.01606649126972765
14	post_wins,0.013664385589914648
15	TSht_x,0.011224949353737294

1	feature,importance
2	rank,0.15090804591901033
3	L,0.15018656252565843
4	win_loss_ratio,0.10826237343932239
5	confW,0.10179850276649119
6	won_x,0.08152249541795509
7	confL,0.06885370775987251
8	lost_x,0.04939940071306944
9	W,0.0489609035026654
10	ORtg,0.027402350109537624
11	post_wins,0.02642808629460465
12	NRtg,0.024496950208131165
13	won_y,0.01873554954978195
14	d_blk,0.01266371189371505
15	DRtg,0.010939174732906275

Some of the Feature Importances for the WE and EA conference models (from left to right)

Annexes

Feature Engineering

Player Efficiency Rating (PER) calculated and saved successfully

	playerID	uPER	PER
0	abrossv01w	0.509929	18.152665
1	abrossv01w	0.491677	17.502921
2	abrossv01w	0.536490	19.098188
3	abrossv01w	0.452381	16.104043
4	abrossv01w	0.491248	17.487662

Player Efficiency Rating table